

Permutation & Combination (Class 11) — The Way You'll Remember for Life **Golden Rule:** Does ORDER matter? If order matters → Permutation. If order does not matter → Combination. Lifetime Memory Trick **P = Position** (Permutation) **C = Collection** (Combination) Important Formulas $nPr = \frac{n!}{(n-r)!}$ $nCr = \frac{n!}{r!(n-r)!}$ $nPr = nCr \times r!$ $n! = n \times (n-1) \times \dots \times 1$ $0! = 1$ When to Use Permutation? Arrangement Ranking Seating Password Captain/Vice-Captain When to Use Combination? Selection Committee Team Formation Choosing Members Real-Life Examples Mobile Password → Permutation Cricket Team Selection → Combination Gold, Silver, Bronze → Permutation WhatsApp Group Members → Combination Easy Practice Questions 1. Arrange 3 books. Answer: $3! = 6$ 2. Select 2 students from 5. Answer: ${}^5C_2 = 10$ 3. Arrange 2 students from 5. Answer: ${}^5P_2 = 20$ 4. Form 3-digit numbers from 1,2,3. Answer: $3! = 6$ 5. Committee of 3 from 8 people. Answer: ${}^8C_3 = 56$ Medium Practice Questions 1. Captain and Vice-Captain from 7 students. Answer: ${}^7P_2 = 42$ 2. Choose 4 players from 10. Answer: ${}^{10}C_4 = 210$ 3. Arrange 4 books from 8. Answer: ${}^8P_4 = 1680$ 4. Seat 5 people in a row. Answer: $5! = 120$ 5. Committee of 4 from 12 people. Answer: ${}^{12}C_4 = 495$ Advanced Practice 1. 4-digit numbers from digits 1–9 without repetition. Answer: ${}^9P_4 = 3024$ 2. Committee of 5 from 20 students. Answer: ${}^{20}C_5 = 15504$ 3. Circular arrangement of 8 people. Answer: $(8-1)! = 5040$ 4. President, Secretary, Treasurer from 10 students. Answer: ${}^{10}P_3 = 720$ 5. Diagonals in a 10-sided polygon. Answer: 35 10-Second Revision 1. Arrangement → Permutation 2. Selection → Combination 3. Order Matters → Permutation 4. Order Doesn't Matter → Combination 5. P = Position 6. C = Collection 7. $nPr = nCr \times r!$ 8. Factorial powers everything